Keilson's theorem and the challenges of time attribution

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Będlewo, May 20, 2026

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Setup and preliminary facts

Let $X = (X_t)_{t\geq 0}$ be a continuous-time birth-death process (BDP) on $\mathbb{N}_0$ (or $[\![0, N]\!] = \{0, \ldots, N\}$ for some $N \geq 1$), with

- positive birth rates b_i for $i = 0, \ldots, N-1$,
- positive death rates d_i for $i = 1, \ldots, N-1$,
- absorbing state N.

We are interested in the law – under $\mathbb{P}_0 = \mathbb{P}[\cdot|X_0 = 0]$ – of the first hitting time of state n, that is, of

$$T_n := \inf\{t \geq 0 : X_t = n\}$$

for $n \in \mathbb{N}$. This requires to look at the spectral properties of the generator of X when killed at T_n, namely

The killed generator

$$L = L_n = \begin{pmatrix} -b_0 & b_0 & 0 & 0 & 0 \\ d_1 & -r_1 & b_1 & 0 & 0 \\ 0 & d_2 & -r_2 & \ddots & \\ & \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & d_{n-1} & -r_{n-1} \end{pmatrix}$$

with $r_i := b_i + d_i$.

It is well-known that $-L$ has n distinct positive eigenvalues

$$\theta_1^{(n)} < \theta_2^{(n)} < \ldots < \theta_n^{(n)},$$

and that the important intertlacement relation

$$\theta_1^{(n)} < \theta_1^{(n-1)} < \theta_2^{(n)} < \theta_2^{(n-1)} < \ldots < \theta_{n-1}^{(n-1)} < \theta_n^{(n)}$$

holds for all $n \in \mathbb{N}$.

The following result is known as Keilson's theorem:

Theorem

Under $\mathbb{P}_0 = \mathbb{P}[\,\cdot\,|\,X_0 = 0]$, *the absorption time* T_n *satisfies*

$$T_n \overset{d}{=} \sum_{k=1}^{n} E_k^{(n)},$$

where $E_1^{(n)}, \ldots, E_n^{(n)}$ *are independent exponentially distributed random variables, more precisely*

$$E_k^{(n)} \overset{d}{=} \mathrm{Exp}(\theta_k^{(n)}) \qquad \text{for } k = 1, \ldots, n.$$

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The proof of these statements depend on an alternative representation of the quantity Var z_n. To this effect, we observe that the Laplace transform of z_n can be factored in the form

$$\varphi_n(s) = \frac{1}{Q_n(-s)} = \frac{1}{\prod_{i=1}^{n}\left(1 + \frac{s}{\alpha_{ni}}\right)} \tag{45}$$

where α_{ni} are the roots of Q_n (recall that the α_{ni} are real and positive).

Why is this result interesting?

Why is this result interesting?

Because it raises the hope that – in the given setup – the eigenvalues of the generator L_n can be given a probabilistic interpretation!

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Answers were given by Fill [5] and by Diaconis & Miclo [2] by using intertwinings and strong stationary duality.

Fix $n \in \mathbb{N}$ and let $\rho = (\rho_i)_{0 \le i \le n}$ be the quasi-stationary distribution of X when killed at n, i.e., if $\mathcal{L}(X_0) = \rho$, then

$$\mathcal{L}(X_t | T_n > t) = \rho \quad \text{for all } t \ge 0,$$

and

$$\mathcal{L}(T_n) = \overbrace{\mathrm{Exp}(\theta_1^{(n)})}^{\text{shows up in Keilson's theorem}}.$$

Recalling that the killed generator $L = L_n$ has n distinct negative eigenvalues

$$-\theta_n^{(n)} < -\theta_{n-1}^{(n)} < \cdots < -\theta_1^{(n)},$$

the quasi-stationary distribution ρ is the normalized left positive eigenvector of $-\theta_1^{(n)}$.

Fix $n \geq 2$ D&M show that there is a dual BDP $X^* = (X^*_t)_{t\geq 0}$ with absorbing state $n-1$ which – owing to work by Fill – can be constructed together with X such that

- $(X_t, X^*_t)_{t\geq 0}$ is a CTMC (Markov coupling) starting at $(0,0)$, and – with $\mathcal{F}^*_t = \sigma(X^*_s, 0 \leq s \leq t)$ –

$$\mathcal{L}(X_t|\mathcal{F}^*_t) = \mathcal{L}(X_t|X^*_t) = \Lambda(X^*_t, \cdot) \quad \text{a.s.}$$

  for a lower triangular Markov kernel Λ, called the link between X and X^*. Algebraic duality relation: $\Lambda L = L^*\Lambda$.

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The D&M approach and strong quasi-stationary times

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  for a lower triangular Markov kernel Λ, called the link between X and X^*. Algebraic duality relation: $\Lambda L = L^*\Lambda$.
- $X_t \leq X^*_t$ for all $t \geq 0$ (follows from the fact that Λ is lower triangular),
- If $T^*_{n-1} = \inf\{t : X^*_t = n-1\}$, then $X_{T^*_{n-1}}$ and T^*_{n-1} are independent and $\mathcal{L}(X_{T^*_{n-1}}) = \rho$.

Regarding the problem of time attribution, this means that the exponential variable with parameter $\theta_1^{(n)}$ can be viewed as the remaining time – namely $T_n - T_{n-1}^*$ – the BDP X needs until absorption at n when reaching the quasi-stationary time distribution at T_{n-1}^*.

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This can be iterated because the dual operator L^* on $[\![0, n-1]\!]$ has eigenvalues

$$-\theta_n^{(n)} < -\theta_{n-1}^{(n)} < \cdots < -\theta_2^{(n)}.$$

In a nutshell: Apply the previous construction to X^* on $[\![0, n-1]\!]$, then to the dual of this dual on $[\![0, n-2]\!]$, and so forth.

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The "terminal" dual, say $\widehat{X}$ – with hitting times $\widehat{T}_k$ – is just a pure birth process with birth rates $\widehat{b}_i = \theta_{n-i}^{(n)}$ for $i = 0, \ldots, n-1$.

We note that $\widehat{T}_0 = 0$ and $\widehat{T}_n = T_n$. Let $\rho^{(k)}$ be the law of $X_{\widehat{T}_k}$ for $k = 0, \ldots, n$, thus

$$\rho^{(0)} = \delta_0, \qquad \rho^{(n-1)} = \rho \qquad \text{and} \qquad \rho^{(n)} = \delta_n.$$

Theorem

The following assertions hold:

- $\mathcal{L}(\widehat{T}_k) = \mathrm{Exp}(\theta_n^{(n)}) * \mathrm{Exp}(\theta_{n-1}^{(n)}) * \cdots * \mathrm{Exp}(\theta_{n-k+1}^{(n)})$.
- *The $\widehat{T}_k$ are strong randomized stopping times, that is, $\widehat{T}_k$ and $X_{\widehat{T}_k}$ are independent.*
- $\widehat{T}_k \le T_k$ *for each $k = 1, \ldots, n$.*
- $\mathcal{L}(X_t | \widehat{T}_k \le t < \widehat{T}_{k+1}) = \rho^{(k)}$ *for all $t \ge 0$ and $k = 1, \ldots, n$, where $\widehat{T}_{n+1} := \infty$.*

Second decomposition of T_n

There is another – obvious – decomposition of T_n into n independent random variables:

Ladder decomposition: $$T_n = \sum_{k=1}^{n} T_{k-1,k},$$

where $T_{k-1,k} := T_k - T_{k-1}$ equals the k-th ladder epoch, i.e., time the process X needs to move from state $k-1$ to state k.

We observe that, by strong Markov property,

$$\mathbb{P}_0[T_{k-1,k} \in \cdot] = \mathbb{P}_{k-1}[T_k \in \cdot].$$

Moreover, T_{k-1} and $T_{k-1,k}$ are independent.

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This shows that the ladder decomposition of T_n is different from the spectral one in Keilson's theorem (unless $n = 1$).

The mixture

$$\alpha\delta_0 + (1-\alpha)\mathrm{Exp}(\beta), \quad \alpha \in [0,1],\ \beta > 0$$

of a point mass at 0 and an exponential distribution is called continuous linear fractional distribution with parameters α, β and abbreviated as $\mathrm{CLF}(\alpha,\beta)$. Its Laplace transform, say $\varphi_{\alpha,\beta}$, has the linear fractional form

$$\varphi_{\alpha,\beta}(t) = \frac{\alpha t + \beta}{t + \beta}$$

which explains the chosen terminology.

Any mixture

$$\mathbb{F} \;=\; \alpha\mathsf{Exp}(\beta_1) + (1-\alpha)\mathsf{Exp}(\beta_2)$$

with $0 < \alpha < 1$ and $\beta_1, \beta_2 > 0$ is the convolution of an exponential distribution with a continuous linear fractional distribution, namely

$$\mathbb{F} \;=\; \mathsf{CLF}\left(\alpha + (1-\alpha)\frac{\beta_2}{\beta_1}, \beta_2\right) * \mathsf{Exp}(\beta_1)$$

when assuming wlog $\beta_2 < \beta_1$. By letting $\alpha \to 0$ (or directly), we further see that

$$\mathsf{Exp}(\beta_2) \;=\; \mathsf{CLF}\left(\frac{\beta_2}{\beta_1}, \beta_2\right) * \mathsf{Exp}(\beta_1).$$

(Infinite divisibility of exponential distributions in the class of continuous linear fractional ones).

For a quick proof consider again the Laplace transform: Let $\widehat{\mathbb{F}}(t)$ be the Laplace transform of $\mathbb{F}$ and put $\gamma = \beta_2/\beta_1 \in (0,1)$ and $\delta := \alpha + (1-\alpha)\gamma$. Then we compute

$$
\begin{aligned}
\widehat{\mathbb{F}}(t) &= \alpha\,\frac{\beta_1}{t+\beta_1} + (1-\alpha)\frac{\beta_2}{t+\beta_2}\\
&= \left(\alpha + (1-\alpha)\frac{\gamma(t+\beta_1)}{t+\beta_2}\right)\frac{\beta_1}{t+\beta_1}\\
&= \left(\delta + (1-\delta)\frac{\beta_2}{t+\beta_2}\right)\frac{\beta_1}{t+\beta_1}.
\end{aligned}
$$

The law of $T_{n-1,n}$

Let φ_n and $\varphi_{n-1,n}$ denote the Laplace transforms of T_n and $T_{n-1,n}$, respectively. Put

$$\gamma_k^{(n)} := \frac{\theta_k^{(n)}}{\theta_k^{(n-1)}} \quad \text{for } 1 \le k < n \le N.$$

Then all $\gamma_k^{(n)}$ are in $(0,1)$, and

$$\varphi_n(t) \;=\; \varphi_{n-1}(t)\varphi_{n-1,n}(t)$$

since $T_n = T_{n-1} + T_{n-1,n}$ and $T_{n-1},\ T_{n-1,n}$ are independent.

Solving for $\varphi_{n-1,n}(t)$ and using Keilson's theorem, we obtain

$$\begin{aligned}
\varphi_{n-1,n}(t) &= \frac{\varphi_n(t)}{\varphi_{n-1}(t)} = \frac{\prod_{k=1}^{n} \frac{\theta_k^{(n)}}{t+\theta_k^{(n)}}}{\prod_{k=1}^{n-1} \frac{\theta_k^{(n-1)}}{t+\theta_k^{(n-1)}}} \\
&= \frac{\prod_{k=1}^{n-1} \gamma_k^{(n)}(t+\theta_k^{(n-1)})}{\prod_{k=1}^{n}(t+\theta_k^{(n)})} \frac{\theta_n^{(n)}}{t+\theta_n^{(n)}} \\
&= \left[\prod_{k=1}^{n-1}\left(\gamma_k^{(n)} + \overline{\gamma}_k^{(n)} \frac{\theta_k^{(n)}}{t+\theta_k^{(n)}}\right)\right] \frac{\theta_n^{(n)}}{t+\theta_n^{(n)}}.
\end{aligned}$$

This shows that $T_{n-1,n}$ is the sum of $n-1$ independent CLF's and one independent exponential, more precisely:

Proposition

For every $n \in \mathbb{N}$,

$$T_{n-1,n} \stackrel{d}{=} \sum_{k=1}^{n-1} C_k^{(n)} + E_n^{(n)} \tag{1}$$

where $C_1^{(n)}, \ldots, C_{n-1}^{(n)}$ and $E_n^{(n)}$ are independent with

$$C_k^{(n)} \stackrel{d}{=} \mathrm{CLF}(\gamma_k^{(n)}, \theta_k^{(n)})$$

and $E_n^{(n)} \stackrel{d}{=} \mathrm{Exp}(\theta_n^{(n)})$ as earlier.

An alternative way of stating (2) is as follows:

$$T_{n-1,n} \stackrel{d}{=} \sum_{k=1}^{n-1} I_k^{(n)} E_k^{(n)} + E_n^{(n)}$$

where all random variables on the r.h.s. are independent, and

$$I_k^{(n)} \stackrel{d}{=} \text{Bern}(\gamma_k^{(n)}), \quad E_k^{(n)} \stackrel{d}{=} \text{Exp}(\theta_k^{(n)}).$$

In the same manner, one can determine the law of $T_{n-m,n} := T_n - T_{n-m}$:

$$T_{n-m,n} \stackrel{d}{=} \sum_{k=1}^{n-m} C_k^{(j)} + \sum_{l=n-m+1}^{n} E_l^{(j)} \tag{2}$$

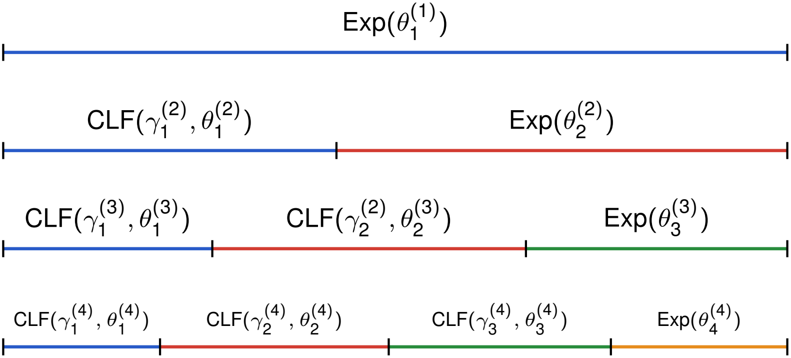
for each $1 \leq m < n$.

With regard to the problem of time attribution, the following consequence of the above proposition is most interesting:

$$E_k^{(k)} + \sum_{j=k+1}^{n} C_k^{(j)} \stackrel{d}{=} \mathrm{Exp}(\theta_k^{(n)}) \tag{3}$$

for each $k = 1, \ldots, n$.

Splittings of $[\![0, T_1]\!]$, $[\![T_1, T_2]\!]$, $[\![T_2, T_3]\!]$, and $[\![T_3, T_4]\!]$:



Third decomposition of T_n

The last decomposition is in terms of the occupation/sojourn times of the visited states. Let

$$S_k^{(n)} = \int_{T_{n-1}}^{T_n} \mathbf{1}_{\{k\}}(X_t)\,dt,$$

$$U_k^{(n)} = \int_0^{T_n} \mathbf{1}_{\{k\}}(X_t)\,dt = \sum_{m=1}^{n} S_k^{(m)}$$

be the times the process X spends in state k before hitting n for the first time when starting from $n-1$ and from 0, respectively. Then

Sojourn time decomposition: $$T_n = \sum_{k=0}^{n-1} U_k^{(n)}.$$

- The $S_k^{(n)}$ are independent for $n = 0, 1, \ldots,$ but they are not for $k = 0, \ldots, n-1$.
- The $U_k^{(n)}$ are not independent, neither for $n = 0, 1, \ldots$ nor for $k = 0, \ldots, n-1$.
- The law of $U_k^{(n)}$ is the convolution of the laws of $S_k^{(m)}$ for $m = 1, \ldots, n$.

To determine the distribution of the occupation times $S_k^{(n)}$, we observe the recursion in distribution

$$T_{n,n+1} \stackrel{d}{=} E_n(0) + I_n(T_{n-1,n}(1) + T_{n,n+1}(1))$$

which upon iteration directly leads to

$$T_{n,n+1} \stackrel{d}{=} \sum_{i=1}^{\eta_n} E_n(i) + \sum_{i=1}^{\eta_n - 1} T_{n-1,n}(i), \tag{4}$$

where

- I_n is a Bernoulli variable with parameter $d_n r_n^{-1}$,
- η_n is a positive geometric variable with parameter $b_n r_n^{-1}$,
- E_n is an exponential variable with parameter r_n,
- $E_n(i)$ and $T_{n,n+1}(i)$ for $i \geq 1$ are copies of E_n and $T_{n,n+1}$, respectively.
- All introduced random variables are independent of all other occurring random variables.

We note the well-known fact

$$\sum_{i=1}^{\eta_n} E_n(i) \overset{d}{=} \mathrm{Exp}\left(\frac{b_n}{r_n} \cdot r_n\right) = \mathrm{Exp}(b_n).$$

- For each $n \geq 1$, let $(Z_k^{(n)})_{k\geq 0}$ be a GWPVE with one ancestor and reproduction law $\mathrm{Geom}_+(b_{n-k-1}/r_{n-k-1})$ in generation $k \in [\![0, n-1]\!]$. (For $k \geq n$, these laws are of no relevance here).
- Each $(Z_k^{(n)})_{k\geq 0}$ and η_n is independent of all other occurring random variables.

We note that

$$Z_k^{(n)} \stackrel{d}{=} \mathrm{Geom}_+\left(\prod_{i=n-k}^{n-1} \frac{b_i}{r_i}\right)$$

for any $n \in \mathbb{N}$ and $k \in [\![1, n]\!]$.

Proposition

The laws of the $S_k^{(n)}$ are given as follows:

$$S_{n-1}^{(n)} \stackrel{d}{=} \sum_{j=1}^{\eta_{n-1}} E_{n-1}(j) \stackrel{d}{=} \mathrm{Exp}(b_{n-1})$$

$$S_0^{(n)} \stackrel{d}{=} \sum_{j=1}^{Z_{n-2}^{(n)}} \sum_{j=1}^{\eta_1(i)} E_0(i,j) \stackrel{d}{=} \mathrm{CLF}\left(\frac{\beta_2^{(n)}-1}{\beta_2^{(n)}}, \frac{\alpha_2^{(n)} b_1 r_1^{-1}}{\beta_2^{(n)}}\right)$$

and, for $k = 1, \ldots, n-2$,

$$S_k^{(n)} \stackrel{d}{=} \sum_{i=1}^{Z_{n-1-k}^{(n)}} \sum_{j=1}^{\eta_k(i)} E_k(i,j) \stackrel{d}{=} \mathrm{CLF}\left(\frac{\beta_{k+1}^{(n)}-1}{\beta_{k+1}^{(n)}}, \frac{\alpha_{k+1}^{(n)} b_k r_k^{-1}}{\beta_{k+1}^{(n)}}\right),$$

Proposition

where $\alpha_n^{(n)} = \beta_n^{(n)} := 1$ *and*

$$\alpha_k^{(n)} := \prod_{i=k}^{n-1} \frac{b_i}{d_i}, \qquad \beta_k^{(n)} := \alpha_k^{(n)} + \beta_{k+1}^{(n)} = \sum_{i=k}^{n} \alpha_i^{(n)}$$

for $k = 1, \ldots, n-1$*.*

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Thank you for your attention.